Sumedh Kothari
skothari67@gatech.edu | linkedin.com/in/sumedh-kothari | github.com/sumedhk0 | Personal Portfolio

EDUCATION

Georgia Institute of Technology **Expected May 2028**
B.S. in Chemical and Biomolecular Engineering GPA: 3.91
Minor in Computer Science – Intelligence
Minor in Physics – Quantum Sciences and Technology

- **Coursework:** Computer Organization and Programming; Data Structures and Algorithms; Differential Equations
- **Honors:** PACESHIP Scholar, TechHack Valley 2026 Hackathon Winner, Symposium of Rising Scholars Presenter, Eagle Scout

EXPERIENCE

Witness AI Mountain View, CA
Machine Learning Intern May – September 2026

- Built an adversarial self-play framework for automated jailbreak discovery that combines generative adversarial training, SFT, and GRPO with verifiable rewards, optimizing attack policies via iterative reward-based learning and Thompson sampling; generated 10,000 jailbreak prompts reaching 60% attack success on Claude, Qwen, and Gemma, used to train a prompt-safety classifier
- Built LiLAC (Living Literature Adversarial Corpus), an end-to-end agentic research pipeline in Python (PyTorch, vLLM, HuggingFace) that autonomously processed 1,000+ arXiv papers, synthesized emerging LLM red-teaming methods, and continuously converted them into novel attack strategies and adversarial data for training and evaluating LLM safety classifiers and jailbreak guards
- Developed a scalable evaluation framework that generated 3,000+ adversarial prompts, benchmarked 8 state-of-the-art open-source and proprietary LLMs, fine-tuned 3 safety classifiers on the synthesized data, and automated customer-facing safety reports

ChemE Cube | Georgia Tech Atlanta, GA
Software & Mechanical Lead August 2025 – Present

- Developed a Gaussian-process Bayesian optimization framework for CO_2 capture experiments that learns a surrogate model of capture performance from experimental data and selects the next high-value operating conditions to test, accelerating reactor optimization
- Lead mechanical development of a $1\,ft^3$ direct air capture reactor for the AIChE ChemE Cube competition (Nov 2026), managing a 5-member subteam through end-to-end CAD, prototyping, and systems integration into a fully fabricated, test-ready prototype
- Designed airflow distribution and sorbent containment subsystems using parametric CAD, 3D printing, and CNC machining, achieving a 75% reduction in ambient CO_2 concentration during bench-scale validation of the full direct air capture prototype

RESEARCH

Ramprasad Laboratory | Georgia Tech Atlanta, GA
Machine Learning Researcher July 2026 – Present

- **PolyFrameNet: Block-Factored Polymer MLIP**
  - Designed a polymer MLIP that factors chains into graph-defined rigid blocks with local coordinate frames, replacing equivariant tensor products with plain MLPs while guaranteeing exact O(3) and permutation symmetry via Weisfeiler–Lehman frame averaging
  - Achieved 2.5–5.5× faster inference and 3.4× less GPU memory than MACE and Allegro at a matched training budget (90 min, one L40S) on 876K-atom OPoly26 data; matched MACE force error on polyethylene (13.7 meV/Å) and scaled to 26K-atom melts
  - Open-sourced the PyTorch/ASE implementation and trained weights with an 81-test correctness suite and a block registry that flags chemistries outside the training set, so users know when predictions extrapolate beyond the chemical blocks seen during training
- **PolyDYN** — Building a multi-agent LLM framework for autonomous polymer design and discovery with MLIP-driven property screening

PROJECTS

PolyT5-RLVR: Polymer Encoder-Decoder Model August – September 2026

- Reproduced the Ramprasad Lab's PolyT5, a 7.5M-parameter T5 encoder-decoder pretrained on 92.3M polymers, and post-trained it with GRPO against chemistry-verifiable rewards (RLVR), cutting the rate of chemically invalid polymer generations by 79%
- Replaced text decoding with a linear regression head for Tg prediction, cutting prediction error 5.2% (28.60 → 27.11 K) and structurally eliminating catastrophic decode failures; the gain held for both seeds on all 10 splits of a 70-run, five-factor ablation
- Added distribution-free split-conformal prediction intervals for Tg, calibrated on a held-out split (±54.35 K at 90% coverage), converting single-point predictions into calibrated uncertainty ranges with a finite-sample marginal coverage guarantee

Machine Learning-Assisted Particle Swarm Optimization March – May 2026

- Awarded a $1,000 PACESHIP Scholar grant to research surrogate-assisted particle swarm optimization (PSO) for efficient black-box optimization, using cheap learned surrogate models to reduce the number of expensive objective-function evaluations
- Built a hybrid black-box optimizer combining Gaussian-process surrogate models, LightGBM-guided repositioning of stagnating particles, and adaptive exploration–exploitation balancing to steer the swarm toward promising regions of the search space
- Developed an autoresearch loop that autonomously ran experiments, tuned hyperparameters, and refined search strategies; improved on baseline PSO by 28% on average across benchmarks (Sphere 97.8%, Rastrigin 99.9%, Ackley 90.4%, Rosenbrock 58.8%)

Panoptic Pigskin: NFL 4D Gaussian Splatting February – September 2026

- Reconstructed NFL plays in 3D from just two broadcast camera angles, validated at 0.15 m median error against official NFL player-tracking data (41× below per-frame calibration), by self-training YOLOv8x-pose per play on 115K SMPL-X pseudo-labels projected across positionally and rotationally solved cameras (Ultralytics YOLOv8, SMPL-X, SMPLest-X, EasyOCR, OpenCV, SciPy)
- Cut physically implausible motion from 17% to 5% of frames by smoothing each track along its camera's depth axis, powering free-viewpoint Gaussian-splat replays and an automated report grading all 22 players (separation, time to throw, air yards, YAC)

SKILLS

- **Programming:** Python, C++, CUDA, Java, MATLAB, Linux
- **Machine Learning:** PyTorch, TensorFlow, HuggingFace Transformers, vLLM, scikit-learn, LightGBM, OpenCV
- **Atomistic Simulation:** ASE, LAMMPS, VASP, MACE, Allegro
- **CAD & Fabrication:** SolidWorks, Fusion, 3D printing, CNC machining
- **Languages:** English, Hindi, Spanish